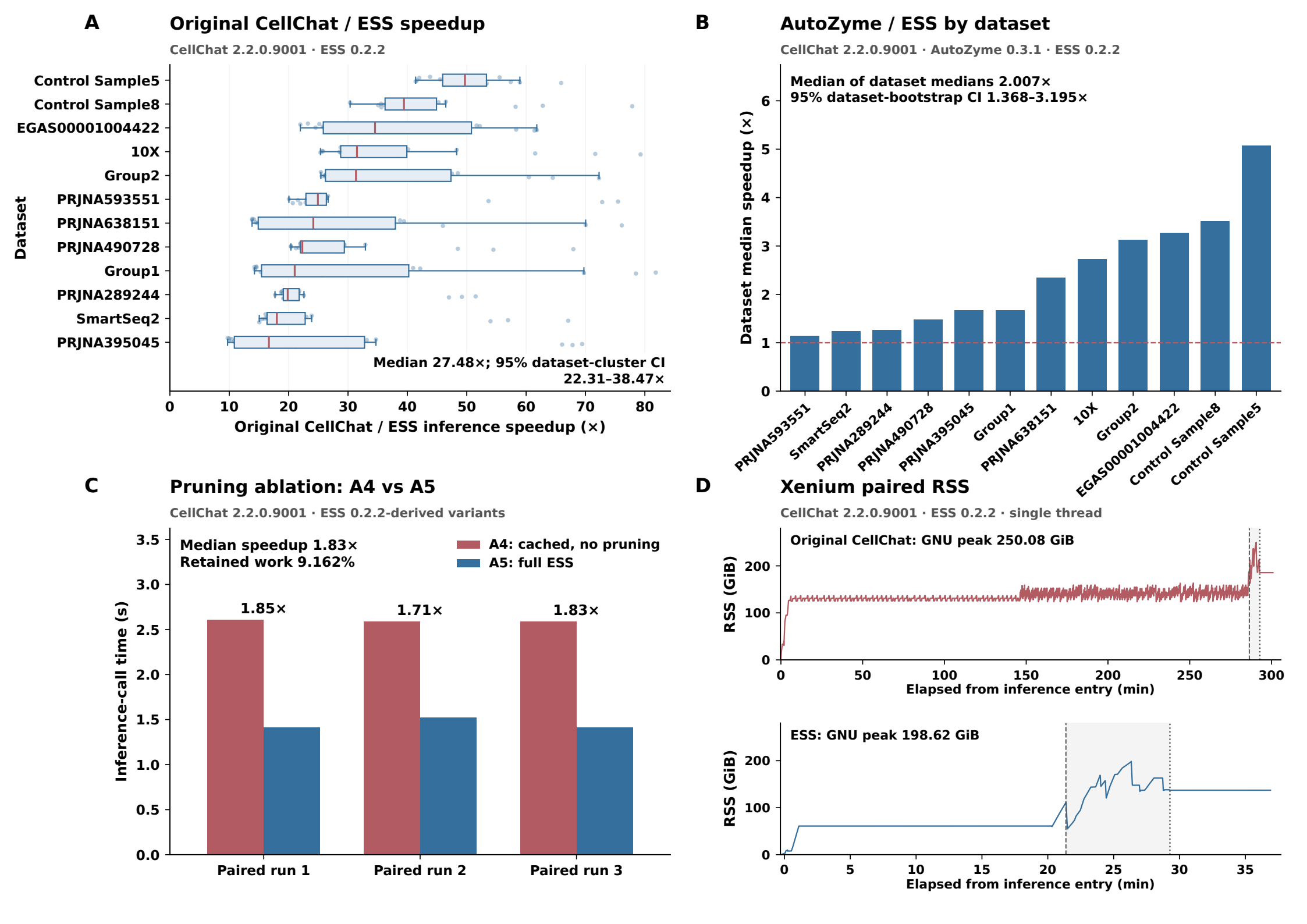
A
Original CellChat / ESS speedup
CellChat 2.2.0.9001 · ESS 0.2.2
Control Sample5
Control Sample8
EGAS00001004422
10X
Group2
PRJNA593551
PRJNA638151
PRJNA490728
Group1
PRJNA289244
SmartSeq2
PRJNA395045
Dataset
Median 27.48×; 95% dataset-cluster CI
22.31–38.47×
0
10
20
30
40
50
60
70
80
Original CellChat / ESS inference speedup (×)
B
AutoZyme / ESS by dataset
CellChat 2.2.0.9001 · AutoZyme 0.3.1 · ESS 0.2.2
Median of dataset medians 2.007×
95% dataset-bootstrap CI 1.368–3.195×
Dataset median speedup (×)
0
1
2
3
4
5
6
PRJNA593551
SmartSeq2
PRJNA289244
PRJNA490728
PRJNA395045
Group1
PRJNA638151
10X
Group2
EGAS00001004422
Control Sample8
Control Sample5
C
Pruning ablation: A4 vs A5
CellChat 2.2.0.9001 · ESS 0.2.2-derived variants
Median speedup 1.83×
Retained work 9.162%
A4: cached, no pruning
A5: full ESS
Inference-call time (s)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
1.85×
1.71×
1.83×
Paired run 1
Paired run 2
Paired run 3
D
Xenium paired RSS
CellChat 2.2.0.9001 · ESS 0.2.2 · single thread
Original CellChat: GNU peak 250.08 GiB
RSS (GiB)
0
100
200
0
50
100
150
200
250
300
Elapsed from inference entry (min)
ESS: GNU peak 198.62 GiB
RSS (GiB)
0
100
200
0
5
10
15
20
25
30
35
Elapsed from inference entry (min)